

Importing & Managing Data Securely in ServiceNow

Problem Statement:

Associating every record with an employee and retrieving relevant employee details, such as the department, to make reporting more convenient.

Objective:

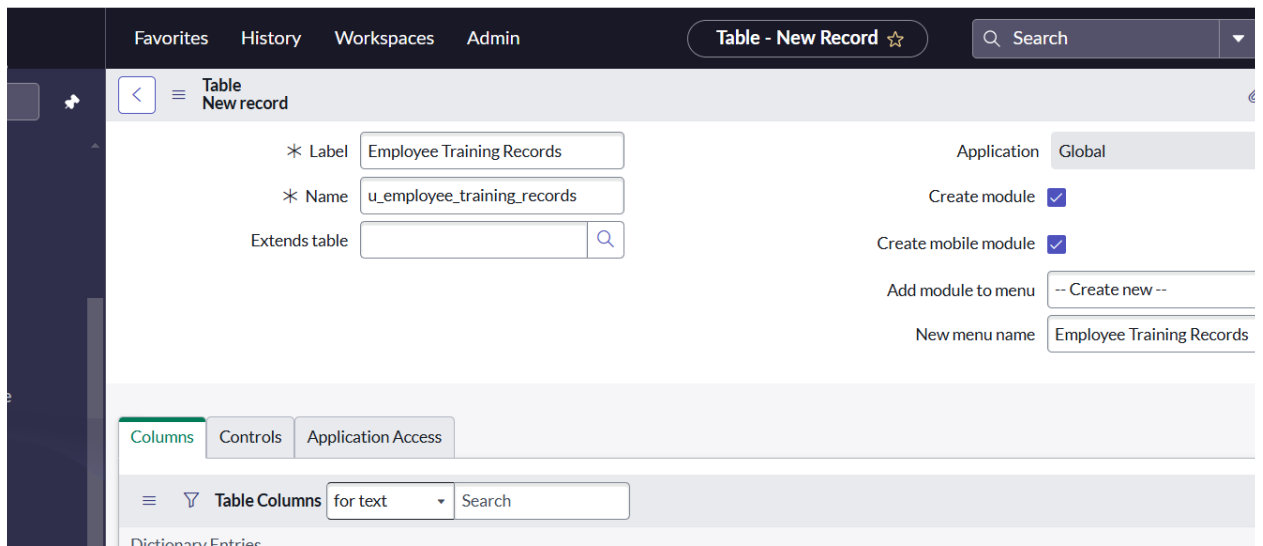
The objective of this initiative is to implement an Access control list by using Impersonate user. This provides a solid foundation for managing and securing data in servicenow using key platform features.

TASK INITIATION

Milestone 1: Tables

Activity 1: Creating a Table

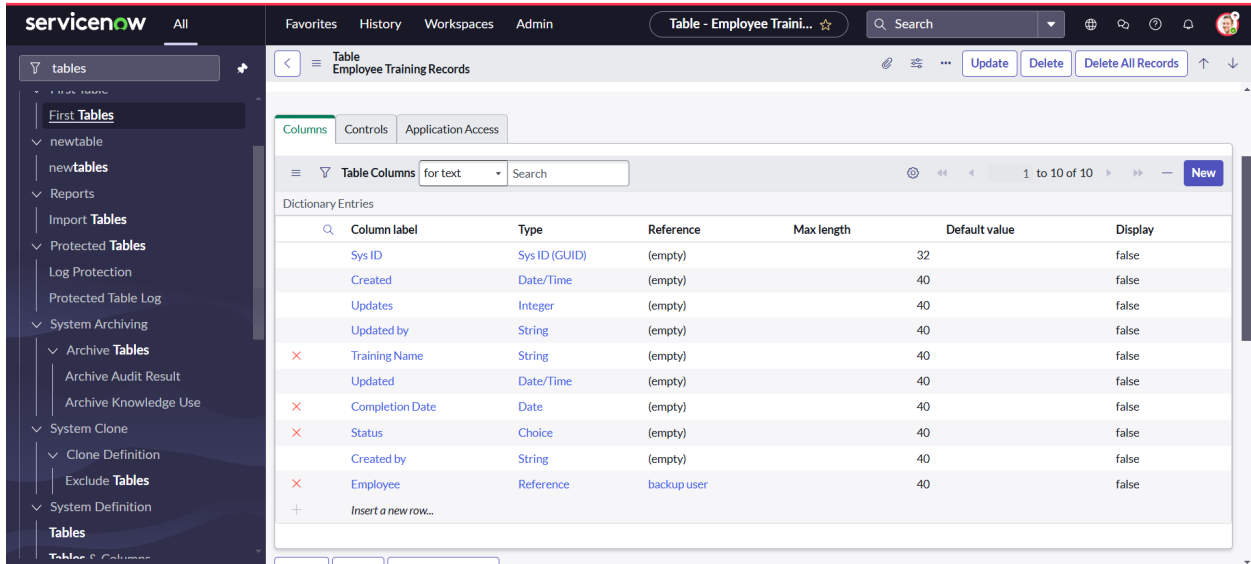
1. Launch ServiceNow.
2. Select All >> search for Tables.
3. Choose Tables under System Definition.
4. Select New.
5. Fill the following details to create a new Table



The screenshot shows the 'Table - New Record' form in ServiceNow. The form is divided into several sections. The top section contains fields for 'Label' (Employee Training Records), 'Name' (u_employee_training_records), and 'Extends table'. The right section contains fields for 'Application' (Global), 'Create module' (checked), 'Create mobile module' (checked), 'Add module to menu' (Create new), and 'New menu name' (Employee Training Records). The bottom section contains tabs for 'Columns', 'Controls', and 'Application Access'. The 'Columns' tab is selected, showing a table with columns for 'Table Columns', 'for text', and 'Search'.

6. Add the following fields in the Columns section:

- Training Name (Type: String)
- Completion Date (Type: Date)
- Status (Type: Choice)
- Employee (Type: Reference), (Reference field to sys_user table)

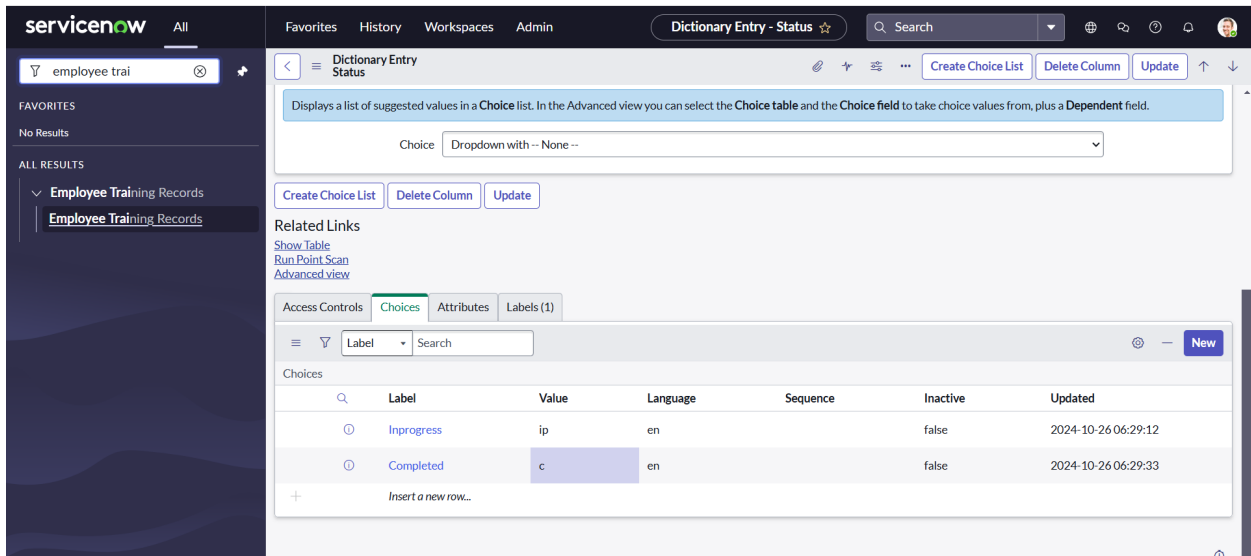


The screenshot shows the ServiceNow interface for editing the 'Employee Training Records' table. The 'Dictionary Entries' tab is active, displaying a list of fields with their properties.

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32		false
Created	Date/Time	(empty)	40		false
Updates	Integer	(empty)	40		false
Updated by	String	(empty)	40		false
Training Name	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Completion Date	Date	(empty)	40		false
Status	Choice	(empty)	40		false
Created by	String	(empty)	40		false
Employee	Reference	backup user	40		false

7. Click Submit.

8. Click on Status Field to add choices in the Dictionary Entry Status



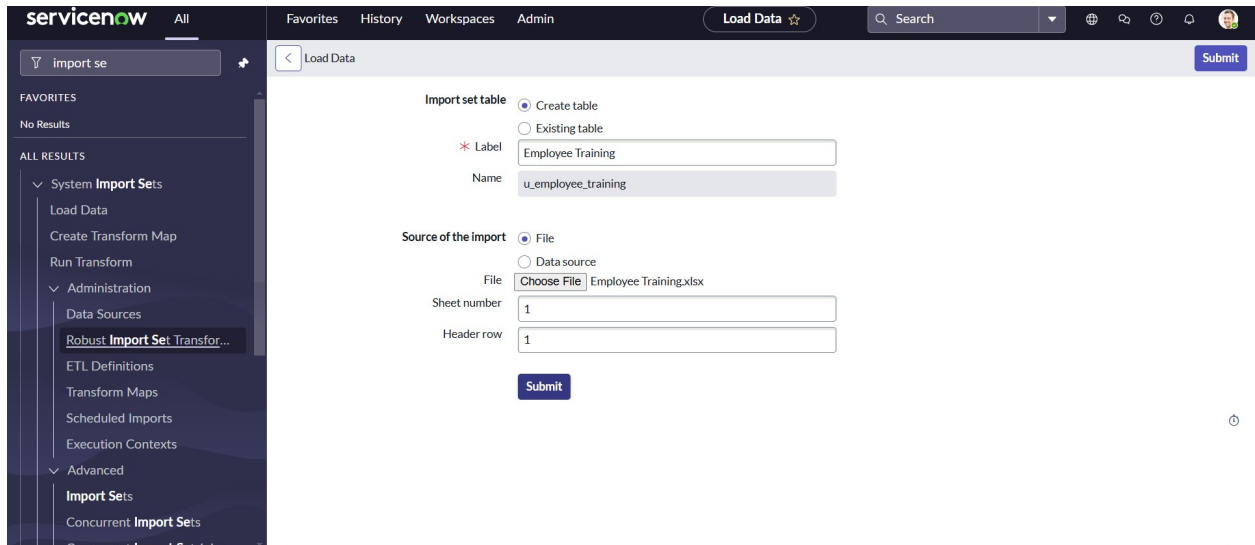
The screenshot shows the ServiceNow interface for the 'Dictionary Entry - Status'. The 'Choices' tab is active, displaying a list of choices for the 'Status' field.

Label	Value	Language	Sequence	Inactive	Updated
Inprogress	ip	en		false	2024-10-26 06:29:12
Completed	c	en		false	2024-10-26 06:29:33

Milestone 2: Import Data

Activity 1: Data Import

1. Prepare an .xlsx spreadsheet with the four fields (Training Name, Completion Date, Status and Employee), then download the file.
2. Launch ServiceNow.
3. Click on All >> System Import Sets.
4. Select Load Data and upload the file prepared with these four fields: Training Name, Completion Date, Status and Employee.
5. Label: Employee Training
6. Name: u_employee_training



The screenshot shows the ServiceNow 'Load Data' interface. On the left is a navigation pane with 'import se' in the search bar. Under 'ALL RESULTS', 'System Import Sets' is expanded, showing 'Load Data' as the selected option. The main area is titled 'Load Data' and contains the following fields:

- Import set table:** Radio buttons for 'Create table' (selected) and 'Existing table'.
- Label:** Text field containing 'Employee Training'.
- Name:** Text field containing 'u_employee_training'.
- Source of the import:** Radio buttons for 'File' (selected) and 'Data source'.
- File:** A 'Choose File' button followed by the text 'Employee Training.xlsx'.
- Sheet number:** Text field containing '1'.
- Header row:** Text field containing '1'.
- Submit:** A blue button at the bottom.

7. Click Submit.

servicenow All

import se

FAVORITES
No Results

ALL RESULTS

- System **Import Sets**
 - Load Data
 - Create Transform Map
 - Run Transform
 - Administration
 - Data Sources
 - Robust Import Set Transfor...**
 - ETL Definitions
 - Transform Maps
 - Scheduled Imports
 - Execution Contexts
 - Advanced
 - Import Sets**
 - Concurrent **Import Sets**
 - Concurrent **Import Set Jobs**

Progress

Name	ImportProcessor
State	Complete
Completion code	Success
Message	Processed: 4, inserts 3, updates 0, errors 0, empty and ignored 1, ignored errors 0 (0:00:00.915)

Next steps...

- [Import sets](#) Go to the import sets for this data load
- [Loaded data](#) Go to the newly imported data inside the staging table: u_employee_training
- [Create transform map](#) Create a transform map for the newly staged data
- [Import log](#) View the import log

Activity 2: Map Fields

1. Open Service Now
2. Click on All >> System Import Sets>> Administration >> Transform Maps
3. Fill the following details to create a new Table

servicenow All

import set

Load Data

Create Transform Map

Run Transform

Administration

- Data Sources
- Robust Import Set Transfor...**
- ETL Definitions
- Transform Maps
- Scheduled Imports
- Execution Contexts
- Advanced
 - Import Sets**
 - Concurrent **Import Sets**
 - Concurrent **Import Set Jobs**
 - Multi **Import Sets**
 - Progress
 - Transform History
 - Transform Errors

Table Transform Map - ...

Table Transform Map
New record

* Name Employee Transform

* Source table Employee Training [u_employ...]

Active ☒

Run business rules ☒

Enforce mandatory fields No

Copy empty fields ☐

Create new record on empty coalesce fields ☐

Application Global

Created

* Target table Employee Training Records [u...]

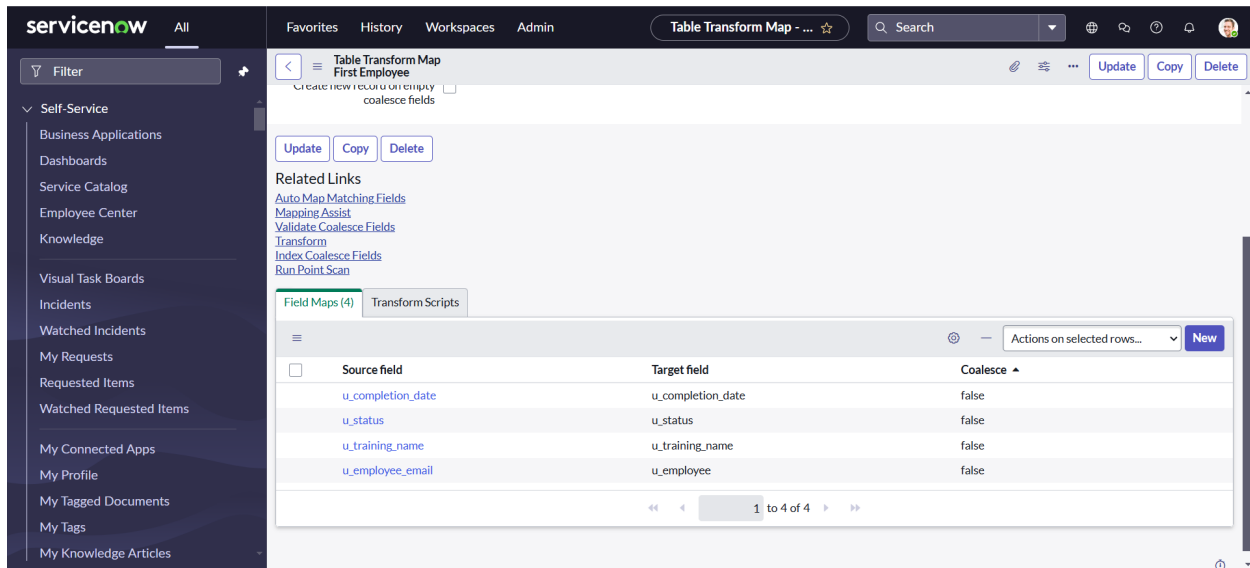
Order 100

Run script ☐

Submit

4. Click on Submit.

5. Add Field Maps as Shown

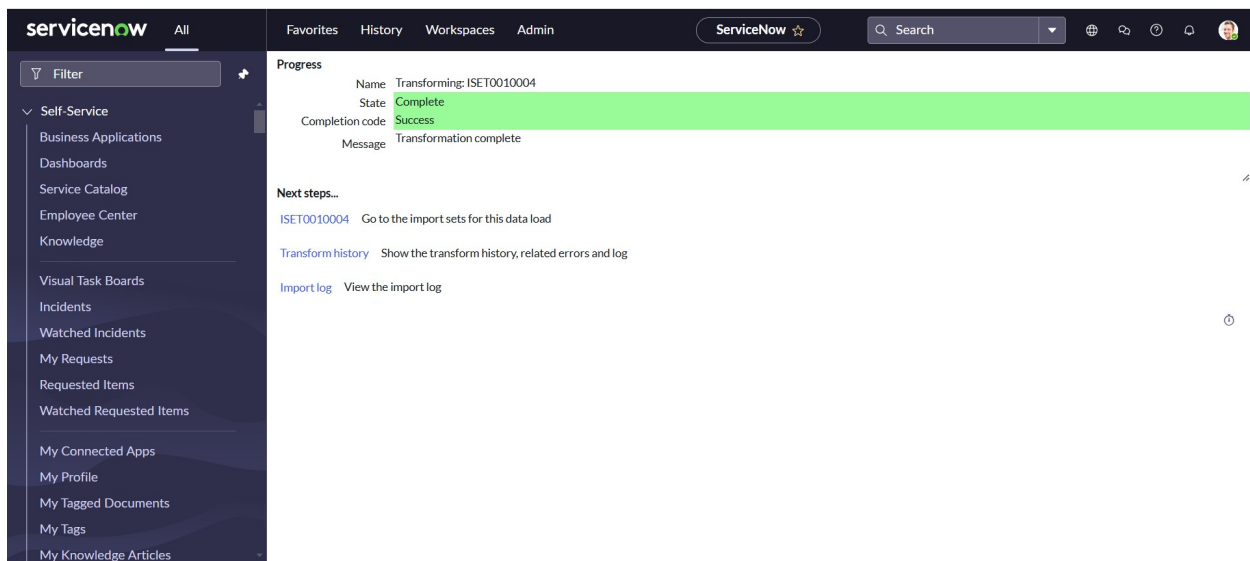


The screenshot shows the ServiceNow 'Table Transform Map' interface for the 'First Employee' table. The 'Field Maps (4)' tab is active, displaying a table with the following data:

Source field	Target field	Coalesce
u_completion_date	u_completion_date	false
u_status	u_status	false
u_training_name	u_training_name	false
u_employee_email	u_employee	false

Below the table, there are navigation controls showing '1 to 4 of 4' records.

6. Under Related Links Click Transform to run the import.



The screenshot shows the ServiceNow 'Progress' page for the transformation job. The job details are as follows:

Name	State	Completion code	Message
Transforming: ISET0010004	Complete	Success	Transformation complete

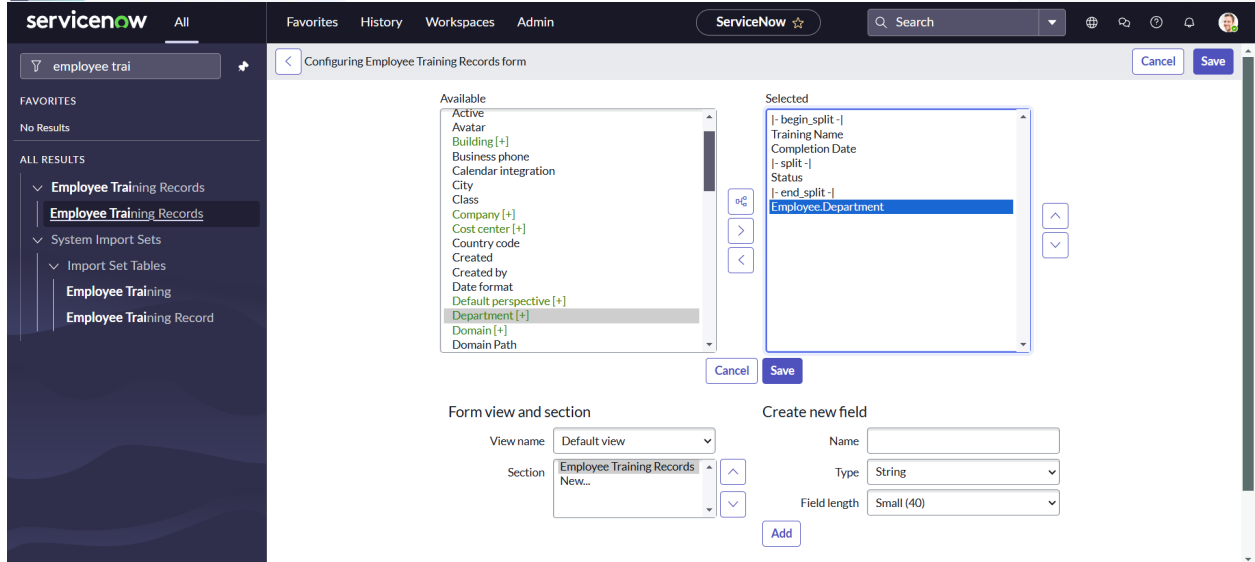
Under 'Next steps...', there are three links:

- [ISET0010004](#) Go to the import sets for this data load
- [Transform history](#) Show the transform history, related errors and log
- [Import log](#) View the import log

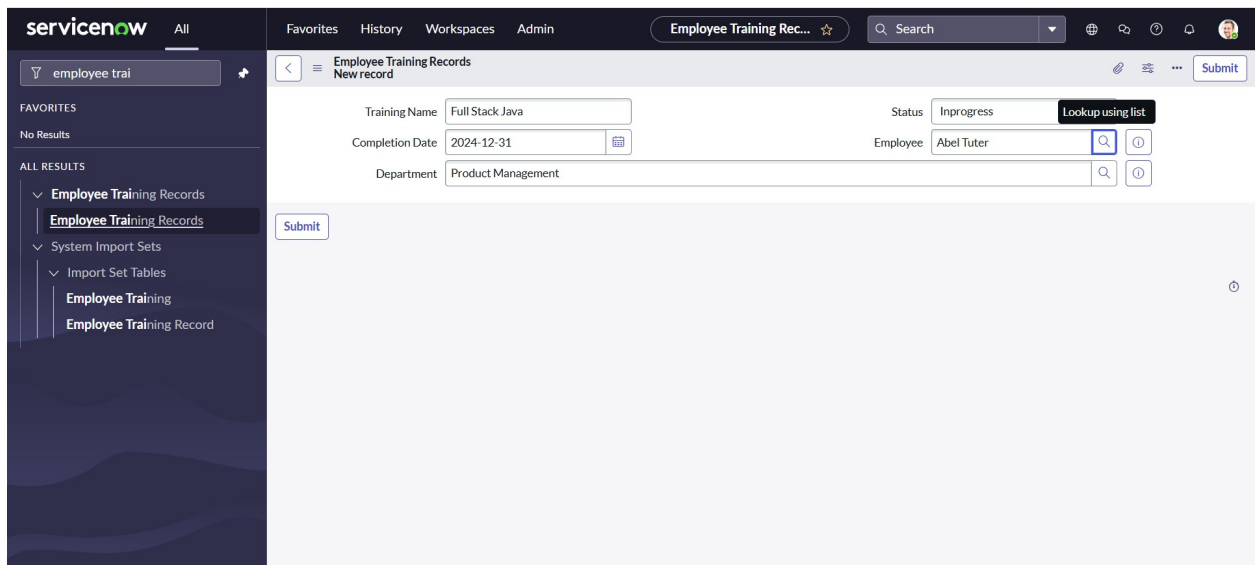
Milestone 3: Accessing Employee Department Information Using Dot-Walking

Activity 1: Using Dot-Walking

1. Launch ServiceNow.
2. Click on All >> Employee Training Records >> Employee Training Records.
3. Open the form by selecting the New button.
4. Open Additional Actions in the left corner and select Configure and Form Layout.
5. Select Employee[+], use dot-walking to choose Department, and move it to the right.
6. Save the changes.



7. Now you can see the field in the List view.



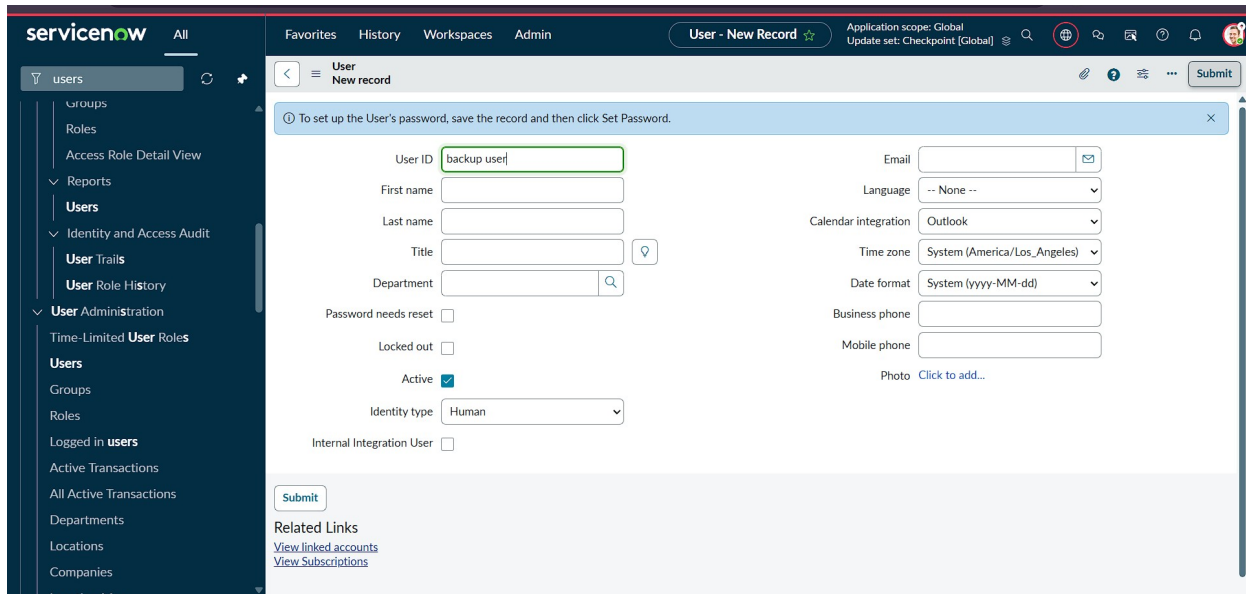
1.

Milestone 4: User and Role Creation

Activity 1: Creating a User

1. Open ServiceNow

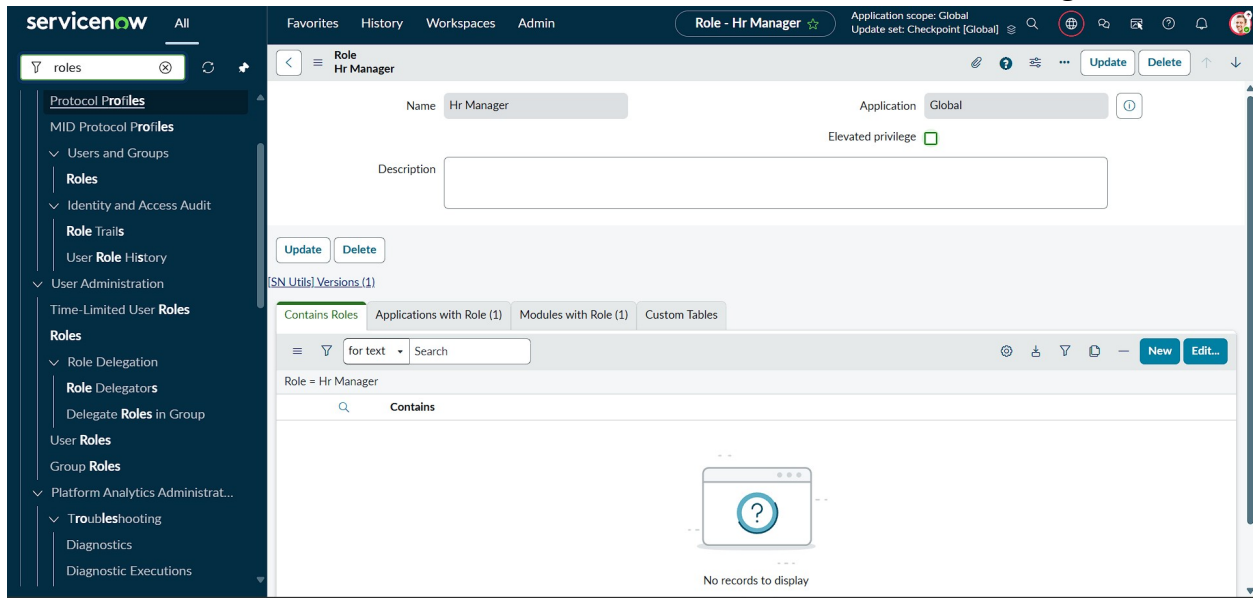
- Click on All>>User Administration>>Users create a new user : backup user



The screenshot shows the ServiceNow 'User - New Record' form. The left sidebar contains a navigation menu with options like 'users', 'Groups', 'Roles', 'Access Role Detail View', 'Reports', 'Users', 'Identity and Access Audit', 'User Trails', 'User Role History', 'User Administration', 'Time-Limited User Roles', 'Users', 'Groups', 'Roles', 'Logged in users', 'Active Transactions', 'All Active Transactions', 'Departments', 'Locations', and 'Companies'. The main form area has a header 'User - New Record' and a sub-header 'New record'. A message at the top states: 'To set up the User's password, save the record and then click Set Password.' The form fields include: User ID (filled with 'backup user'), First name, Last name, Title, Department, Password needs reset (checkbox), Locked out (checkbox), Active (checkbox, checked), Identity type (dropdown, set to 'Human'), Internal Integration User (checkbox), Email, Language (dropdown, set to '-- None --'), Calendar integration (dropdown, set to 'Outlook'), Time zone (dropdown, set to 'System (America/Los_Angeles)'), Date format (dropdown, set to 'System (yyyy-MM-dd)'), Business phone, Mobile phone, and Photo (link to add...). A 'Submit' button is at the bottom left. Related links 'View linked accounts' and 'View Subscriptions' are at the bottom right.

Activity 2: Creating a Role

- Open ServiceNow
- Click on all>>User Administration>>Roles>> create a new role : Hr Manager

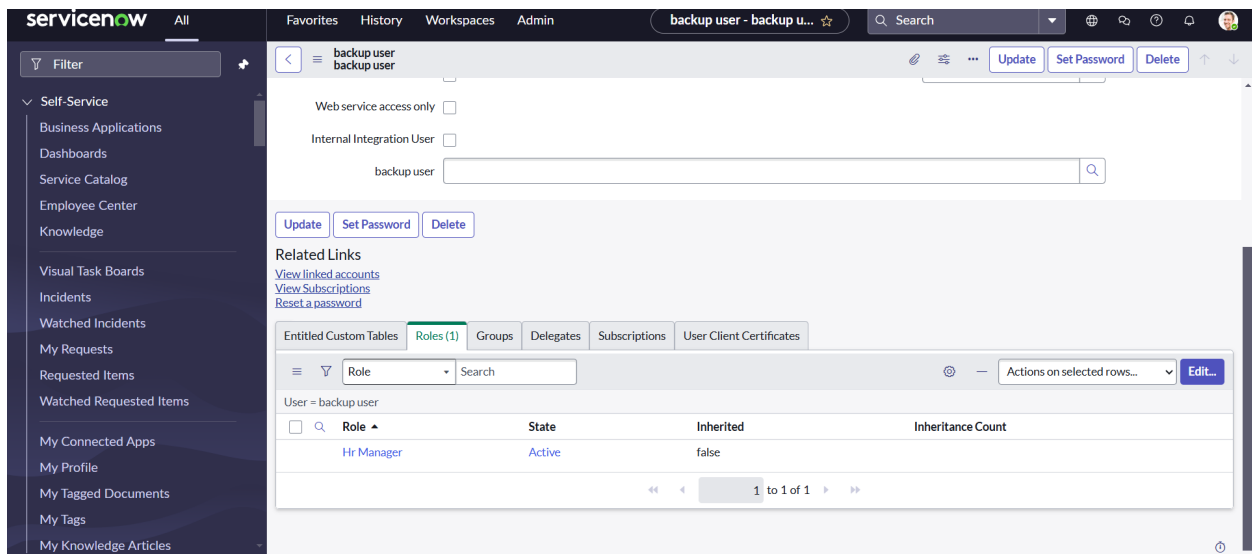


The screenshot shows the ServiceNow 'Role - Hr Manager' form. The left sidebar contains a navigation menu with options like 'roles', 'Protocol Profiles', 'MID Protocol Profiles', 'Users and Groups', 'Roles', 'Identity and Access Audit', 'Role Trails', 'User Role History', 'User Administration', 'Time-Limited User Roles', 'Roles', 'Role Delegation', 'Role Delegators', 'Delegate Roles in Group', 'User Roles', 'Group Roles', 'Platform Analytics Administrat...', 'Troubleshooting', 'Diagnostics', and 'Diagnostic Executions'. The main form area has a header 'Role - Hr Manager' and a sub-header 'Hr Manager'. The form fields include: Name (filled with 'Hr Manager'), Application (dropdown, set to 'Global'), Elevated privilege (checkbox), and Description. There are 'Update' and 'Delete' buttons. Below the form, there is a section 'Contains Roles' with a search bar and a table. The table has a header 'Role = Hr Manager' and a column 'Contains'. The table is empty, and a message at the bottom states 'No records to display'.

Activity 3: Assigning the Role to a User

- Open ServiceNow

2. Click All>>Users (under User Definition) >>open the backup user.
3. Under roles click Edit and add the HR Manager Role.

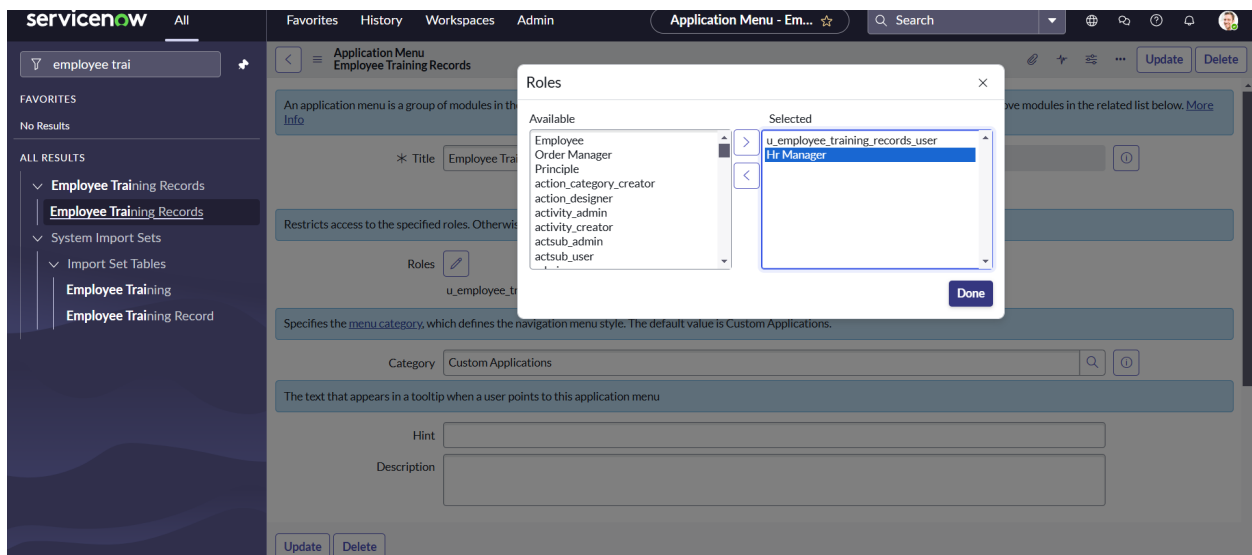


The screenshot shows the ServiceNow interface for configuring a 'backup user'. The left sidebar contains navigation links like 'Self-Service', 'Business Applications', 'Dashboards', 'Service Catalog', 'Employee Center', 'Knowledge', 'Visual Task Boards', 'Incidents', 'Watched Incidents', 'My Requests', 'Requested Items', 'Watched Requested Items', 'My Connected Apps', 'My Profile', 'My Tagged Documents', 'My Tags', and 'My Knowledge Articles'. The main content area shows the 'backup user' configuration with options for 'Web service access only' and 'Internal Integration User'. Below these are 'Update', 'Set Password', and 'Delete' buttons. A 'Related Links' section includes 'View linked accounts', 'View Subscriptions', and 'Reset a password'. At the bottom, there is a table titled 'Entitled Custom Tables' with a tab for 'Roles (1)'. The table shows one role, 'Hr Manager', which is 'Active' and 'Inherited' from 'false'.

Role	State	Inherited	Inheritance Count
Hr Manager	Active	false	1 to 1 of 1

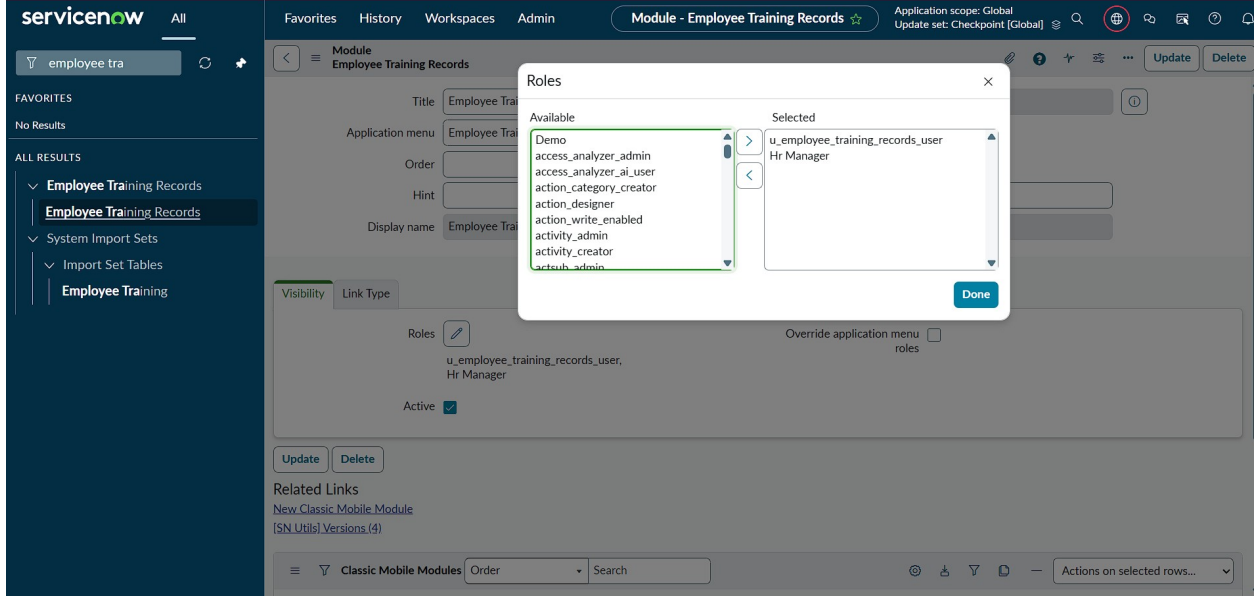
Activity 4: Adding Roles to the Application Menu and Modules

1. Open ServiceNow
2. Click All>>Employee Training Records (click the edit icon in the side).



The screenshot shows the ServiceNow interface for configuring an 'Application Menu'. The left sidebar contains navigation links like 'employee trai'. The main content area shows the 'Application Menu - Employee Training Records' configuration. A 'Roles' dialog box is open, showing a list of available roles and a selected role, 'u_employee.training_records_user'. The dialog box has a 'Done' button. The background configuration page includes fields for 'Title', 'Roles', 'Category', 'Hint', and 'Description'.

3. Click All>>Employee Training Records>>Employee Training Records (click the edit icon in the side).

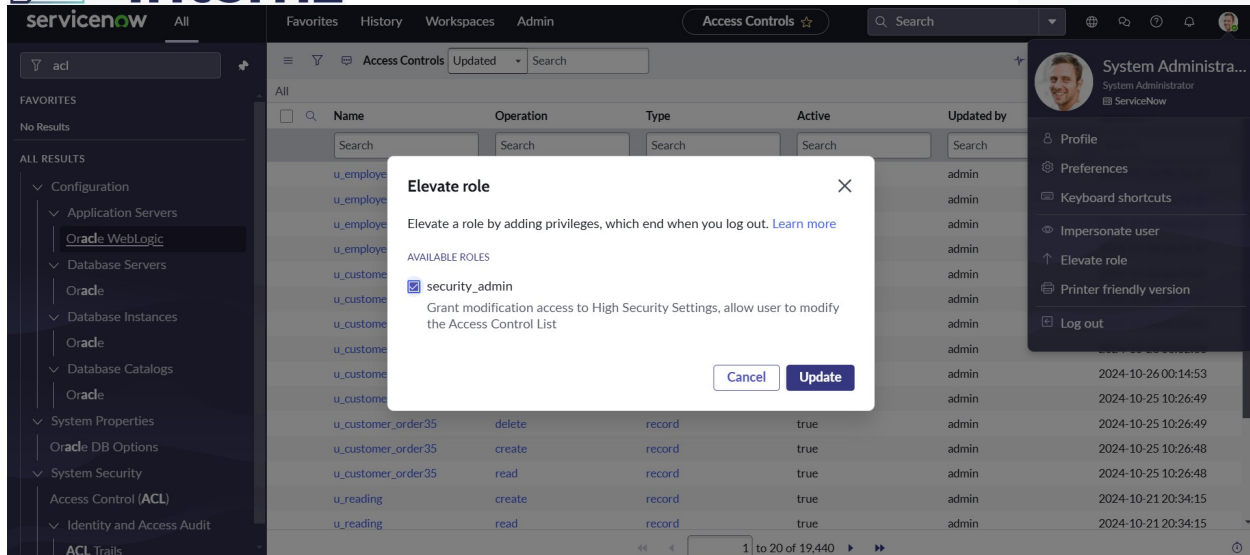


The screenshot shows the ServiceNow interface for the 'Employee Training Records' module. A 'Roles' dialog box is open, allowing the user to select roles for the application menu. The 'Available' list includes roles like 'Demo', 'access_analyzer_admin', and 'access_analyzer_ai_user'. The 'Selected' list currently contains 'u_employee_training_records_user' and 'Hr Manager'. The background shows the module configuration page with fields for Title, Application menu, Order, Hint, and Display name. The 'Roles' field is currently set to 'u_employee_training_records_user, Hr Manager'.

Milestone 5: Access Control Lists (ACL)

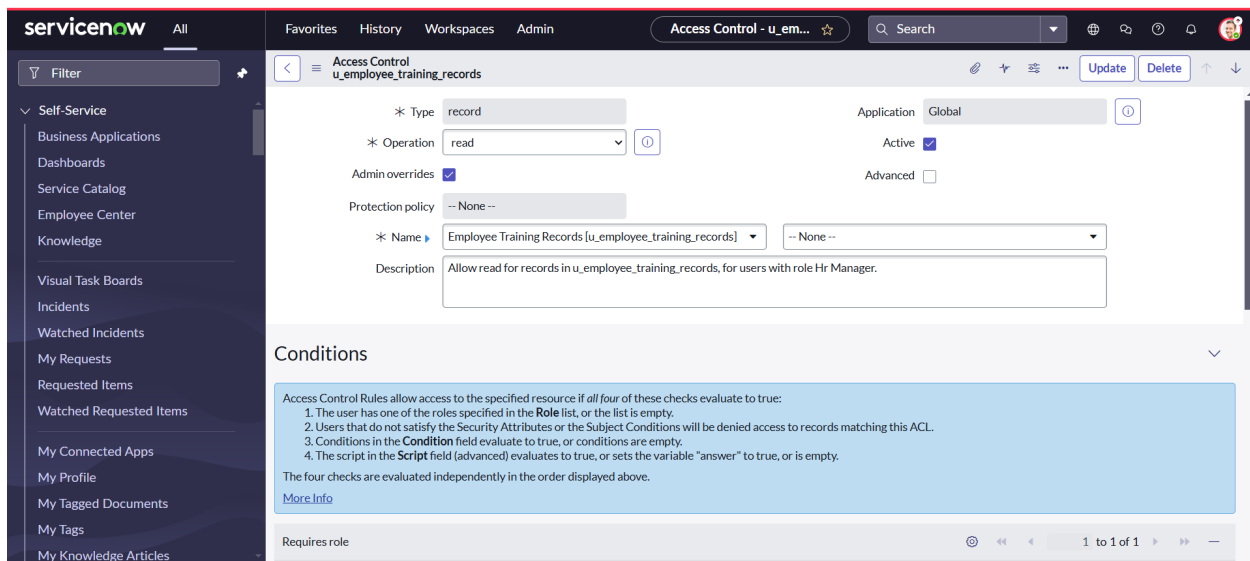
Activity 1: Updating the Role

1. Open ServiceNow
2. Click on Profile>>Elevate Role.
3. Select the Security_admin option and update the record.



Activity 2: Creating ACLs

1. Create a new ACL and provide Read access to the Employee Training Records table.



2. Assign the HR Manager role to the ACL.

servicenow All

Access Control - u_em...

Search

Filter

Self-Service

- Business Applications
- Dashboards
- Service Catalog
- Employee Center
- Knowledge
- Visual Task Boards
- Incidents
- Watched Incidents
- My Requests
- Requested Items
- Watched Requested Items
- My Connected Apps
- My Profile
- My Tagged Documents
- My Tags
- My Knowledge Articles

Access Control u_employee_training_records

Description: Allow read for records in u_employee_training_records, for users with role Hr Manager.

Conditions

Access Control Rules allow access to the specified resource if all four of these checks evaluate to true:

1. The user has one of the roles specified in the **Role** list, or the list is empty.
2. Users that do not satisfy the Security Attributes or the Subject Conditions will be denied access to records matching this ACL.
3. Conditions in the **Condition** field evaluate to true, or conditions are empty.
4. The script in the **Script** field (advanced) evaluates to true, or sets the variable "answer" to true, or is empty.

The four checks are evaluated independently in the order displayed above.

[More Info](#)

Requires role

Role
Hr Manager
Insert a new row...

Security Attribute Condition

Local or Existing ☐ Existing ☒ Local

3. Create another ACL and repeat the same process to provide Write access.

servicenow All

Access Control - u_em...

Search

Filter

Self-Service

- Business Applications
- Dashboards
- Service Catalog
- Employee Center
- Knowledge
- Visual Task Boards
- Incidents
- Watched Incidents
- My Requests
- Requested Items
- Watched Requested Items
- My Connected Apps
- My Profile
- My Tagged Documents
- My Tags
- My Knowledge Articles

Access Control u_employee_training_records

* Type: record

* Operation: write

Application: Global

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None --

* Name: Employee Training Records [u_employee_training_records]

Description: Allow write for records in u_employee_training_records, for users with role Hr Manager.

Conditions

Access Control Rules allow access to the specified resource if all four of these checks evaluate to true:

1. The user has one of the roles specified in the **Role** list, or the list is empty.
2. Users that do not satisfy the Security Attributes or the Subject Conditions will be denied access to records matching this ACL.
3. Conditions in the **Condition** field evaluate to true, or conditions are empty.
4. The script in the **Script** field (advanced) evaluates to true, or sets the variable "answer" to true, or is empty.

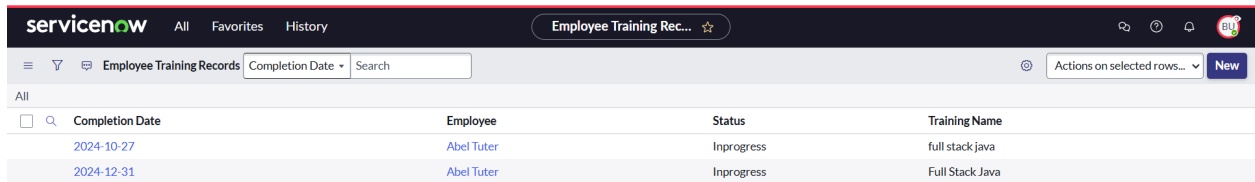
The four checks are evaluated independently in the order displayed above.

[More Info](#)

Requires role

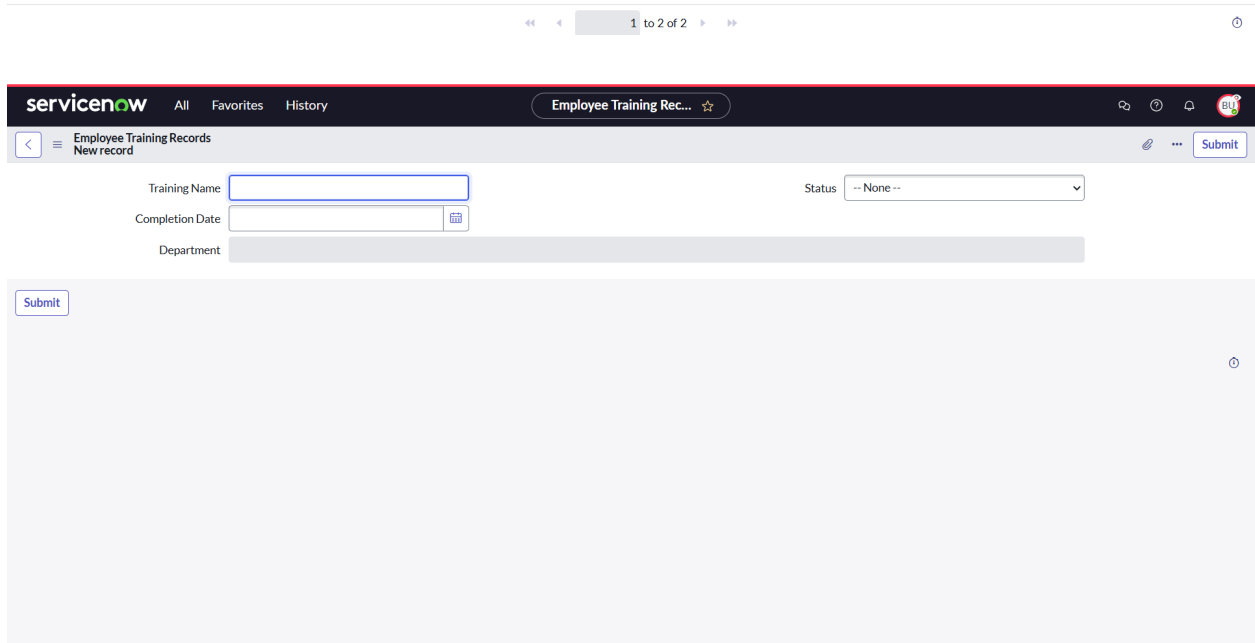
Activity 1: Testing the Results

1. Impersonate the sys_user and Search Employee Training Records
2. Now You can see and edit the Fields



The screenshot shows the ServiceNow interface for 'Employee Training Records'. The table has four columns: Completion Date, Employee, Status, and Training Name. There are two records listed.

Completion Date	Employee	Status	Training Name
2024-10-27	Abel Tutor	Inprogress	full stack.java
2024-12-31	Abel Tutor	Inprogress	Full Stack Java



The screenshot shows the ServiceNow interface for 'Employee Training Records' in form view. The form includes fields for Training Name, Completion Date, Department, and Status. The Status field is currently set to '-- None --'. There is a 'Submit' button at the bottom left.

Training Name:

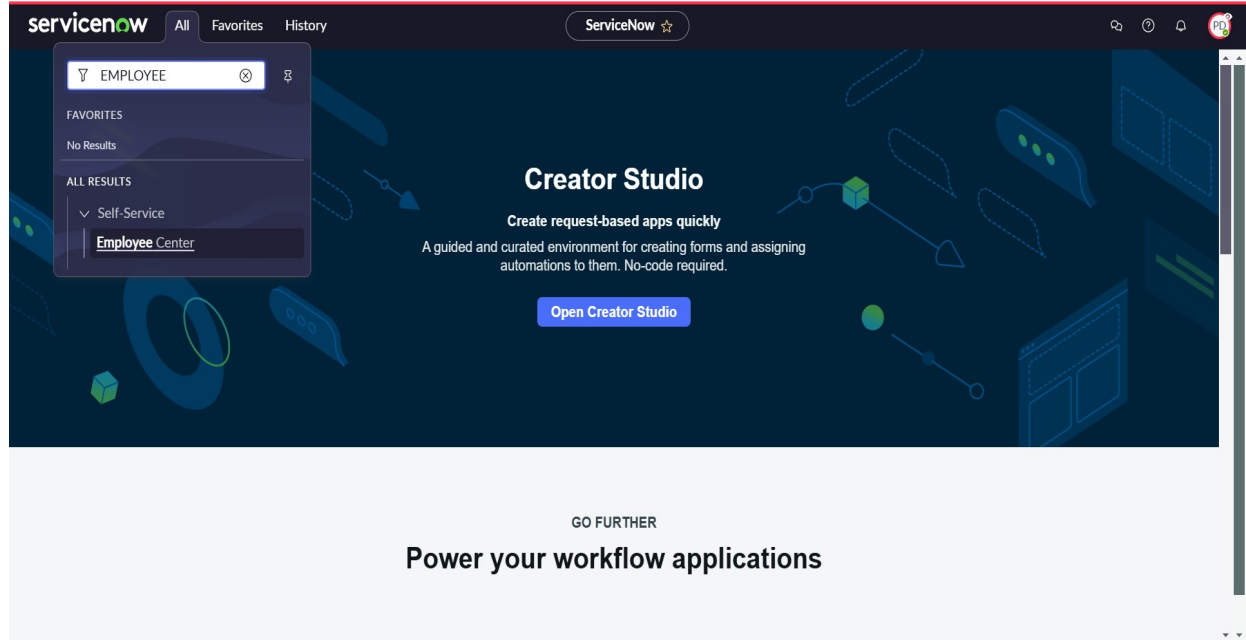
Completion Date:

Department:

Status: -- None --

Submit

3. Impersonate the other User you cannot See the Table



Conclusion :

This project demonstrated the main process of importing data into ServiceNow through Import Sets, using dot-walking to access related employee information efficiently, and applying Access Control Rules (ACLs) to strengthen data security. By combining these features, the project supported accurate data integration, organized relationships between records, and protection of sensitive information through role-based access control.